20260922 Quiz

for (MATH301) Modern Algebra I

Jungwoo Kim
Department of Mathematics
Pohang University of Science and Technology

Exercise 1.1.21. Let G be a finite group and let x be an element of G of order n. Prove that if n is odd, then $x = (x^2)^k$ for some integer k.

Solution. Since n is odd, $k = (n+1)/2$ is an integer. As $x^n = 1_G$,

$$(x^2)^k = x^{n+1} = x^n x = x.$$

Thus x is a power of x^2.

□

Exercise 1.1.25. Prove that if $x^2 = 1_G$ for every $x \in G$, then G is abelian.

Solution. The hypothesis gives $x^{-1} = x$ for every $x \in G$. In particular, for any $a, b \in G$,

$$ab = (ab)^{-1} = b^{-1}a^{-1} = ba.$$

Therefore every two elements commute and G is abelian.

□

Exercise 1.1.31. Prove that every finite group of even order has an element of order 2.

Hint. Let $t(G) = \{g \in G \mid g \neq g^{-1}\}$. Pair each member with its inverse, then consider the remaining elements.

Solution. Let $T = \{g \in G \mid g \neq g^{-1}\}$. The elements of T can be paired as $\{g, g^{-1}\}$, with distinct members, so $|T|$ is even. Since $|G|$ is even, $G \setminus T$ also has even cardinality. It contains 1_G, hence contains another element $x \neq 1_G$. By the definition of T we have $x = x^{-1}$, so $x^2 = 1_G$. As x is not the identity, its order is exactly 2.

□

Exercise 1.1.34. If x has infinite order in G, prove that x^n for $n \in \mathbb{Z}$ are all distinct.

Solution. If $x^m = x^n$ for integers $m \neq n$, multiplication by $x^{-\min(m,n)}$ yields $x^{|m-n|} = 1_G$. This is a positive power of x, contradicting its infinite order. Hence $m = n$.

□

Exercise 1.2.3. Use the presentation

$$D_n = \langle r, s \mid r^n = s^2 = 1,\ rs = sr^{-1} \rangle$$

to prove that every element not in $\langle r \rangle$ has order 2. Deduce that D_n is generated by s and sr, both of order 2.

Notation. Under MATH000, D_n denotes the symmetries of a regular n-gon (order $2n$); the textbook also uses the alternative notation D_{2n}.

Solution. Every element outside $\langle r \rangle$ has the form sr^i. From $r^i s = sr^{-i}$,

$$(sr^i)^2 = sr^i sr^i = s(sr^{-i})r^i = 1_G.$$

Such an element is not the identity, so its order is 2. In particular s and sr have order 2. Since $s(sr) = r$, the subgroup generated by s and sr contains both standard generators r, s, and therefore equals D_n.

□

Exercise 1.2.6. Let $x, y \in G$ have order 2, and set $t = xy$. Prove $tx = xt^{-1}$. If $n = |xy| < \infty$, show that x, t satisfy the dihedral relations.

Solution. Since $x^{-1} = x$ and $y^{-1} = y$, we have $t^{-1} = yx$. Therefore

$$tx = (xy)x = x(yx) = xt^{-1}.$$

If $|t| = n < \infty$, then $t^n = x^2 = 1_G$ and $tx = xt^{-1}$. These are precisely the relations satisfied by r, s in the usual dihedral presentation, with $r = t$ and $s = x$.

□

Exercise 1.2.7. Show that $\langle a, b \mid a^2 = b^2 = (ab)^n = 1 \rangle$ is a presentation of D_n, under $a = s$ and $b = sr$.

Hint. Show in both directions that the generators in one presentation can be expressed through the other and satisfy the required relations.

Solution. Starting with the standard generators r, s of D_n, put $a = s$ and $b = sr$. Then $a^2 = 1$, while

$$b^2 = srsr = r^{-1}r = 1, \qquad ab = s(sr) = r,$$

so $(ab)^n = 1$. Moreover, $s = a$ and $r = ab$, so a, b generate D_n.

Conversely, in the group given by the displayed presentation, define $s = a$ and $r = ab$. Then $s^2 = 1$, $r^n = 1$, and

$$rs = (ab)a = aba = a(ba) = sr^{-1},$$

because $a^{-1} = a$ and $b^{-1} = b$. The changes of generators $a = s$, $b = sr$ and $s = a$, $r = ab$ are inverse to each other. Thus the two presentations describe the same group.

□

Exercise 1.2.15. Find a set of generators and relations for the additive group $\mathbb{Z}/n\mathbb{Z}$.

Solution. The class $[1]_n$ generates the additive group, since every class equals $k[1]_n$ for an integer k. Its order is exactly n, so the defining relation is $n[1]_n = [0]_n$. Thus a presentation in multiplicative notation is

$$\mathbb{Z}/n\mathbb{Z} \cong \langle x \mid x^n = 1 \rangle.$$

Every word in the single generator reduces to a power x^k, and the relation identifies precisely exponents congruent modulo n, so no additional relations are needed.

□

Exercise 1.3.5. Find the order of (1 12 8 10 4)(2 13)(5 11 7)(6 9).

Solution. The four cycles are disjoint and have lengths $5, 2, 3, 2$. Their powers become the identity simultaneously exactly when the exponent is divisible by all four lengths. Hence the order is

$$\operatorname{lcm}(5, 2, 3, 2) = \boxed{30}.$$

□

Exercise 1.3.13. Show that an element has order 2 in S_n if and only if its disjoint cycle decomposition is a nonempty product of commuting 2-cycles.

Solution. If σ has order 2, then $\sigma^2 = 1$. The length of every nontrivial disjoint cycle must divide 2; hence each has length 2. At least one occurs because $\sigma \neq 1$. Conversely, a nonempty product of disjoint transpositions has square 1 because its factors commute and each has square 1. It is not the identity, so its order is exactly 2.

□

Exercise 1.3.14. Let p be prime. Show that an element has order p in S_n if and only if its disjoint cycle decomposition is a nonempty product of p-cycles. Give a counterexample for composite p.

Solution. Suppose $|\sigma| = p$. Then $\sigma^p = 1$, so every cycle in the disjoint cycle decomposition must have length dividing p. Since p is prime, each nontrivial cycle has length p. At least one such cycle occurs because $\sigma \neq 1$.

Conversely, a nonempty product of disjoint p-cycles has pth power equal to the identity, since the cycles commute. It is not the identity, so its order is exactly p.

For a composite order the conclusion need not hold: $\sigma = (1\ 2)(3\ 4\ 5\ 6)$ has order $\operatorname{lcm}(2, 4) = 4$, but its decomposition contains a 2-cycle, not only 4-cycles.

□

Exercise 1.3.15. Prove that the order of a permutation equals the least common multiple of its disjoint cycle lengths.

Hint. Use the fact that disjoint cycles commute and that an m-cycle has order m.

Solution. Write $\sigma = c_1 \cdots c_r$ as a product of disjoint nontrivial cycles of lengths $m_1, \ldots, m_r$. Disjoint cycles commute, and a power σ^k is the identity precisely when each c_i^k is the identity: on the support of c_i, all other cycles act trivially. Since $c_i^k = 1$ if and only if $m_i \mid k$, the least positive such k is

$$|\sigma| = \operatorname{lcm}(m_1, \ldots, m_r).$$

For the identity permutation, which has no nontrivial cycles, its order is 1 (equivalently, include its 1-cycles).

□

Exercise 1.3.18. Find all positive integers n for which S_5 contains an element of order n.

Hint. Apply the least-common-multiple formula from Exercise 1.3.15 to all partitions of 5.

Solution. A cycle decomposition partitions the five symbols into disjoint cycles. Apart from fixed points, the possible cycle-length patterns are 2, $2+2$, 3, $3+2$, 4, and 5. The corresponding least common multiples are $2, 2, 3, 6, 4, 5$. Together with the identity (order 1), the possible orders are

$$\boxed{1, 2, 3, 4, 5, 6}.$$

Every listed value is realized by the corresponding cycle pattern, and no other pattern is possible. □

Exercise 1.6.4. Prove that the multiplicative groups $\mathbb{R} \setminus \{0\}$ and $\mathbb{C} \setminus \{0\}$ are not isomorphic.

Solution. The complex number i has multiplicative order 4. An isomorphism preserves element orders. But a real number satisfying $x^4 = 1$ must be ± 1, of orders 1 and 2, respectively. Thus the real multiplicative group has no element of order 4, and the groups cannot be isomorphic. □

Exercise 1.6.14. For a homomorphism $\varphi : G \to H$, show that $\ker \varphi = \{g \in G \mid \varphi(g) = 1_H\}$ is a subgroup of G, and that φ is injective if and only if $\ker \varphi = \{1_G\}$.

Solution. We have $\varphi(1_G) = 1_H$, so $1_G \in \ker \varphi$. If $a, b \in \ker \varphi$, then

$$\varphi(ab^{-1}) = \varphi(a)\varphi(b)^{-1} = 1_H,$$

so $ab^{-1} \in \ker \varphi$. Thus $\ker \varphi \leq G$ by the subgroup criterion.

If φ is injective, then $\varphi(g) = 1_H = \varphi(1_G)$ implies $g = 1_G$. Conversely, if $\ker \varphi = \{1_G\}$ and $\varphi(a) = \varphi(b)$, then $\varphi(ab^{-1}) = 1_H$, so $ab^{-1} = 1_G$ and $a = b$.

□

Exercise 1.6.19. Let $G = \{z \in \mathbb{C} \mid z^n = 1 \text{ for some } n \in \mathbb{N}\}$. For a fixed integer $k > 1$, prove that $\varphi : G \to G$, $z \mapsto z^k$, is a surjective homomorphism but not an isomorphism.

Solution. Since complex multiplication is commutative, $(zw)^k = z^k w^k$, so φ is a homomorphism. Given $w \in G$, choose a complex kth root z of w. As $w^n = 1$ for some $n \geq 1$, we have $z^{kn} = 1$, so $z \in G$ and $\varphi(z) = w$. Thus φ is surjective.

However, for $\zeta = e^{2\pi i/k}$ we have $\zeta \neq 1$, $\zeta \in G$, and $\varphi(\zeta) = 1 = \varphi(1)$. Therefore φ is not injective and hence not an isomorphism.

□

Exercise 1.6.22. Let A be an abelian group and $k \in \mathbb{Z}$. Prove that $a \mapsto a^k$ is a homomorphism $A \to A$. Show that for $k = -1$ it is an automorphism.

Solution. For commuting $a, b \in A$ and any integer k, the exponent laws give $(ab)^k = a^k b^k$. Hence $\varphi_k(ab) = \varphi_k(a)\varphi_k(b)$. For $k = -1$, the map sends a to a^{-1}, and applying it twice returns a. Thus $\varphi_{-1}^{-1} = \varphi_{-1}$, proving that it is an automorphism.

□

Exercise 1.6.25. Let n be a positive integer, let r, s be the usual generators of D_n, and let $\theta = 2\pi/n$.

(a) Prove that

$$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

is the matrix of the counterclockwise rotation of the xy-plane about the origin by θ radians.

(b) Prove that the map $\varphi\colon D_n \to \mathrm{GL}_2(\mathbb{R})$ defined on generators by

$$\varphi(r) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}, \qquad \varphi(s) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

extends to a homomorphism.

(c) Prove that the homomorphism φ in part (b) is injective.

Solution. Put $\mathbf{R} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$ and $\mathbf{S} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.

(a) Rotation through θ maps the standard basis vectors to $(\cos\theta, \sin\theta)^T$ and $(-\sin\theta, \cos\theta)^T$. These vectors form the columns of $\mathbf{R}$, giving the claimed matrix.

(b) The matrices are invertible, with $\mathbf{R}^n = \mathbf{I}$ and $\mathbf{S}^2 = \mathbf{I}$. Direct multiplication yields

$$\mathbf{RS} = \begin{pmatrix} -\sin\theta & \cos\theta \\ \cos\theta & \sin\theta \end{pmatrix} = \mathbf{SR}^{-1}.$$

They therefore satisfy the defining relations of D_n. The presentation gives a homomorphism extending the prescribed values on r, s.

(c) Every element of D_n has the form r^j or sr^j for $0 \le j < n$. The equality $\mathbf{R}^j = \mathbf{I}$ occurs for this range only when $j = 0$, since $\mathbf{R}$ rotates by $2\pi/n$. On the other hand,

$$\det(\mathbf{SR}^j) = (-1)(1) = -1 \neq \det \mathbf{I}.$$

Hence no sr^j lies in the kernel. Thus the kernel is $\{1_G\}$, so the homomorphism is injective.

□